PERFORMANCE SPECIFICATION SHEET

TEMPERATURE REGULATOR[1] FOR BATTERY, RECHARGEABLE, REPAIRABLE, NICKEL CADMIUM, VENTED, 24-VOLT, BB-432B/A

This specification is approved for use within Army CECOM Life Cycle Management Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-81757.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the wiring harness shall be M81757/20-1.

2. Wiring harness assembly. The harness shall fit inside the M81757/19-1 battery and shall consist of the following:

a. One auxiliary receptacle (J2), conforming to the interface requirements of an MS3474W12-10S receptacle, to provide connection to battery sensing and control points. The J2 receptacle shall be connected as shown in Figure 2. Contacts shall be so designed that sockets will not be damaged during repeated mating of counterpart connectors. The back of the connector shall be potted to prevent migration of electrolyte.

b. One normally closed temperature switch (S1) mounted on a removable link. Dimensions for the link shall be equivalent to M81757/3-1 except as modified by Figure 1. Potting shall be employed to isolate the thermoswitch from any unconnected battery voltage. The temperature switch shall have normally closed contacts, which open at 60 ± 2.8°C and reclose at 48.8 ± 2.8°C. The temperature switch shall be certified to withstand a minimum of 10,000 operating cycles at 30VDC and 1.0 ampere.

c. One temperature sensor (transducer) shall be mounted on a removable link with dimensions equivalent to M81757/3-3. Potting shall be employed to isolate the transducer from any unconnected battery voltage. Transducer temperature and resistance profile shall be as specified below.

[1] The Temperature Regulator for the BB-432B/A shall be referred to herein as the “wiring harness” to standardize on specification naming practices.

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Temperature	Acceptable Resistance Range (ohms)
10°C (50°F)	1800-2400
16°C (60°F)	1350-1700
21°C (70°F)	1050-1300
27°C (80°F)	800-1000
32°C (90°F)	650-775
38°C (100°F)	500-625
43°C (110°F)	450-500

d. The removable links used for device mounting shall be copper with a cross-sectional area not less than 0.031 in^2, with nickel plating that complies with SAE-AMS-QQ-N-290, Class 1, Grade E. Finish shall be smooth and free of corrosion, pits, and blow-holes.

e. Four each copper, ring terminal wire lugs, for use with 20 AWG copper wire and #8-32 UNC 3A screws. Lugs shall be located as required in Figure 1.

f. All wire used in the wiring harness shall have insulation with the following properties:

1) Capable of working in environments over 194°F (90°C)
2) Resistant to moisture
3) Alkali resistant
4) Resistant to damage from chafing due to machine vibration
5) Allows the adherence of potting substances

The insulation shall be free from cracks, delaminating of the insulation, splinters, blisters and other non-homogeneities visible to the normal eye. When used, sheathing or heat-shrink shall provide for enough flexibility to allow harness installation in a BB-432B/A battery. Connections requiring potting shall consist of the J2 connector, transducer, thermoswitch, and wire lugs (ring terminals). Potting compound used shall not support flame, absorb moisture, and shall be impervious to the electrolyte of the battery. Potting compound shall be free of gel particles, ingredient lumps, agglomerates, and all foreign substances, such as grit, oil, dirt, or water. Wire colors, sizes, and connection point terminations shall be as follows:

REF LTR	COLOR	AWG (MIN)	TERMINATIONS	
A	Red	20	J2 Socket A	Power (+) Lug
B	Yellow	20	Transducer	Power (+) Lug
C	Yellow	20	J2 Socket C	Transducer
D	Blue	20	J2 Socket D	Thermoswitch
E	Blue	20	J2 Socket E	Thermoswitch
F	White	20	J2 Socket F	Cell # 10 Lug
G	Black	20	J2 Socket J	Power (-) Lug

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3. MIL-PRF-81757 variance. The following exceptions to MIL-PRF-81757 apply.

3.1 Modify the following paragraph:

3.6.2 Fasteners. After the phrase "…free of any coating…" add the following: "except as otherwise specified herein."

3.2 Add the following new paragraphs

"3.6.4.5 Transducer. The harness assembly shall utilize a temperature sensor device that exhibits the resistance-temperature characteristics specified herein. Potting employed to isolate the transducer from any battery voltage shall conform to paragraph 2f herein."

"3.6.4.6 Device mounting. The devices (thermoswitch and transducer) shall be on removable links (connector strap). Devices shall be mounted in such a manner that they shall not become detached during handling. The devices shall be positioned on the links in such a manner that they shall not interfere with the tool operations required for removal and insertion of a cell."

"3.8 Coloring and marking. Delete in its entirety and replace with the following: "For wiring harnesses, the thermoswitch shall be marked S1 and the transducer shall be marked TH1. Wires shall be marked according to the letter designations of paragraph 2f herein."

3.3 Modify the following paragraph:

3.9 Environmental requirements. Following any test requirement, wiring harnesses shall comply with the requirements specified in subparagraphs a, d, f, and i. In addition, the wire insulation shall meet the criteria specified in this specification sheet, paragraph 2f.

3.4 Add the following new paragraph:

"3.10.31 Testing of wiring harness. The harness assembly, when subjected to the connection continuity of harness test, wiring harness temperature test, temperature shock test, or alkali resistance test, shall be as specified herein:

"a. Continuity. When tested as specified in 4.5.32, continuity results shall be as follows:

(1) Test points of 4.5.32a: Continuity

(2) Test point of 4.5.32b: Resistance of paragraph 2c herein

"b. There shall be no evidence of alkali migration into harness wires and components; breakdown of insulation; stripping of metal plating from connectors; corrosion of metal parts; nor cracking of potting compound.

"c. The transducer shall meet the requirements herein.

"d. The thermoswitch shall close or open within the specified temperature ranges herein.

"e. There shall be no mechanical failure of components.

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"f. There shall be no dimensional distortion beyond specified limits."

VERIFICATION

1. MIL-PRF-81757 variance. The following exceptions to MIL-PRF-81757 apply.

1.1 Modify the following paragraphs:

4.2.1 Inspection of batteries. Delete in its entirety and replace with: "For harness-only qualification, two harnesses shall be provided; one for the tests in Table IV of MIL-PRF-81757 and the other for those tests added to table IV by the variance herein."

4.5.3 Electrolyte resistance. Add the following at the end of the paragraph: "Functional testing shall consist of the connection continuity of harness test of 4.5.32."

4.5.16 Vibration test. The wiring harness shall be installed in a M81757/19-1 or similar configuration battery (i.e., M81757/7-2 or M81757/7-3) for vibration testing. No charge or discharge shall be performed before, during, or after vibration testing (i.e., the condition of the cells used will not impact test outcome). Following vibration testing, the wiring harness shall be removed from the battery. Each harness shall be subject to the connection of continuity of harness test at 4.5.32. Visually examine each harness. Harnesses shall meet the requirements of 3.9 as modified herein and 3.10.31.

4.5.17 Temperature shock test. Subject wiring harnesses to the same temperature shock method specified for batteries, except that all battery-specific steps (i.e., cover removal/replacement, charging, discharging) are not applicable. After the temperature shock test, perform the connection continuity of harness test of 4.5.32. Harnesses shall meet the requirements of 3.9 as modified herein and 3.10.31.

1.2 Add the following new paragraphs:

"4.5.32 Connection continuity of harness. Continuity of the wiring harness terminations shall be checked with an ohmmeter between the points specified below. Resistance readings shall comply with 3.10.31.

"a.	J2 Socket A	to	Power (+) Lug
	J2 Socket D	to	J2 Socket E
	J2 Socket F	to	Cell # 10 Lug
	J2 Socket J	to	Power (-) Lug
"b.	J2 Socket C	to	Power (+) Lug (resistance of transducer; see 2c herein)

"4.5.33 Wiring harness temperature test. The following test shall be performed. After completion of testing, the wiring harnesses shall meet the requirements of 3.1 and 3.10.31."

"a. Fixture the wiring harness so that an ohmmeter is connected across the terminals of the transducer and the S1 thermoswitch.

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“b. Place the harness into an environmental chamber; set the chamber to reach the first temperature set point in the table below. Once the set temperature has been reached, allow the test sample to stabilize for not less than 1 hour.

“c. Record resistance values for the transducer and the thermoswitch.

“d. Repeat steps b and c for the second set point in the table below.

“e. Repeat step d until readings have been obtained through set point 4.

“f. Set the chamber to reach a temperature within a range of 56 ± 1.1°C (133°F); allow the test sample to stabilize for not less than 1 hour.

“g. Set the chamber to ramp up the temperature to not greater than 62.8°C (145°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitch S1 continuously.

“h. Once the thermoswitch opens, record the temperature and stop the temperature increase.

“i. Set the chamber to reach a temperature within a range of 50.6 ± 1.1°C (123°F). Allow the test sample to stabilize for not less than 1 hour.

“j. Set the chamber to ramp down the temperature to not less than 46°C (115°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitch S1 continuously.

“k. Once the thermoswitch closes, record the temperature and proceed to the next step.

“l. Return chamber to room ambient conditions. Allow the test sample to stabilize for not less than 1 hour. Remove the test sample and examine wiring harness for the requirements of 3.10.31.

“m. Perform the connection continuity test of 4.5.32.

“Temperature Set Points, BB-432B/A Wiring Harness

(1) 10.0 ± 1.1°C (50°F)
(2) 21.0 ± 1.1°C (70°F)
(3) 32.0 ± 1.1°C (90°F)
(4) 43.0 ± 1.1°C (110°F)”

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1.3 Modify the Tables of MIL-PRF-81757 as follows:

Table IV. Qualification inspection of components and materials. Add the following:

Part	Examinations and tests	Requirement paragraph	Test paragraph
Wiring harness	Connection continuity of harness	3.10.31	4.5.32
Wiring harness	Vibration	3.9	4.5.16
Wiring harness	Temperature shock	3.9, 3.10.31	4.5.17
Wiring harness	Wiring harness temperature test	3.10.31	4.5.33

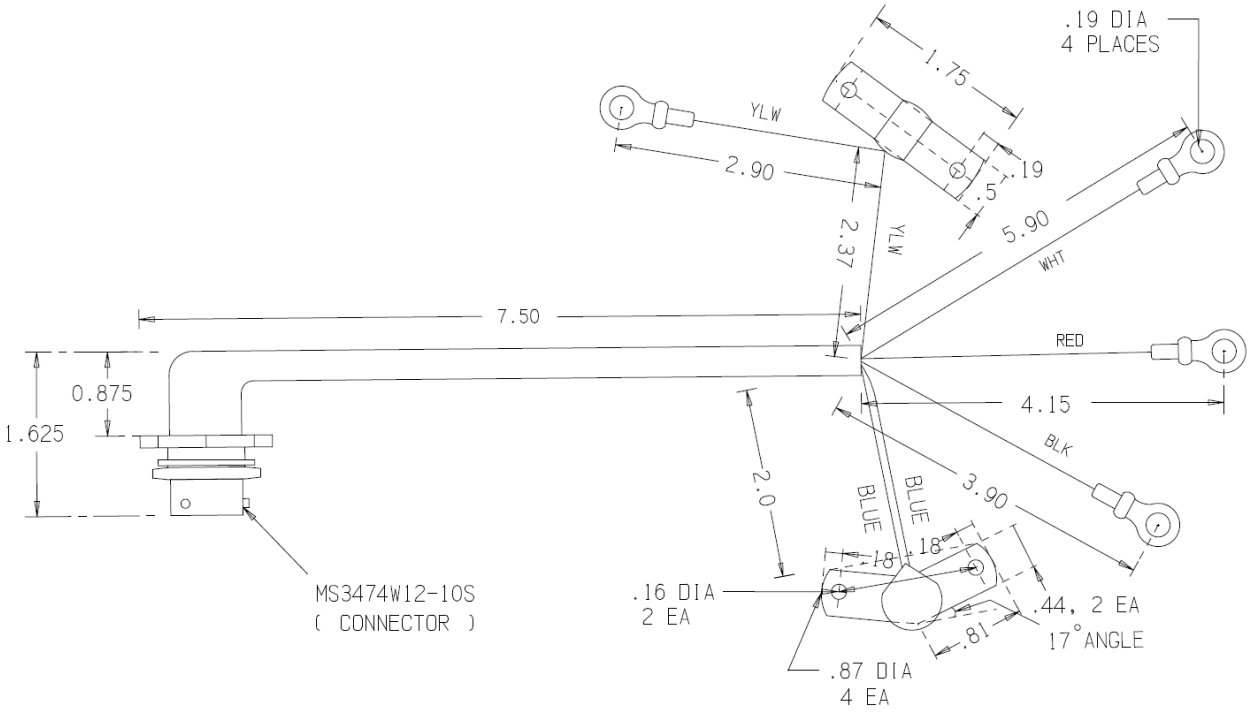
Table VII. Group B inspection. Add the following:

Test No.	Examinations and tests	Requirement paragraph	Test paragraph
	Subgroup I		
I-4	Dielectric strength and insulation resistance	3.9, 3.10.27	4.5.27

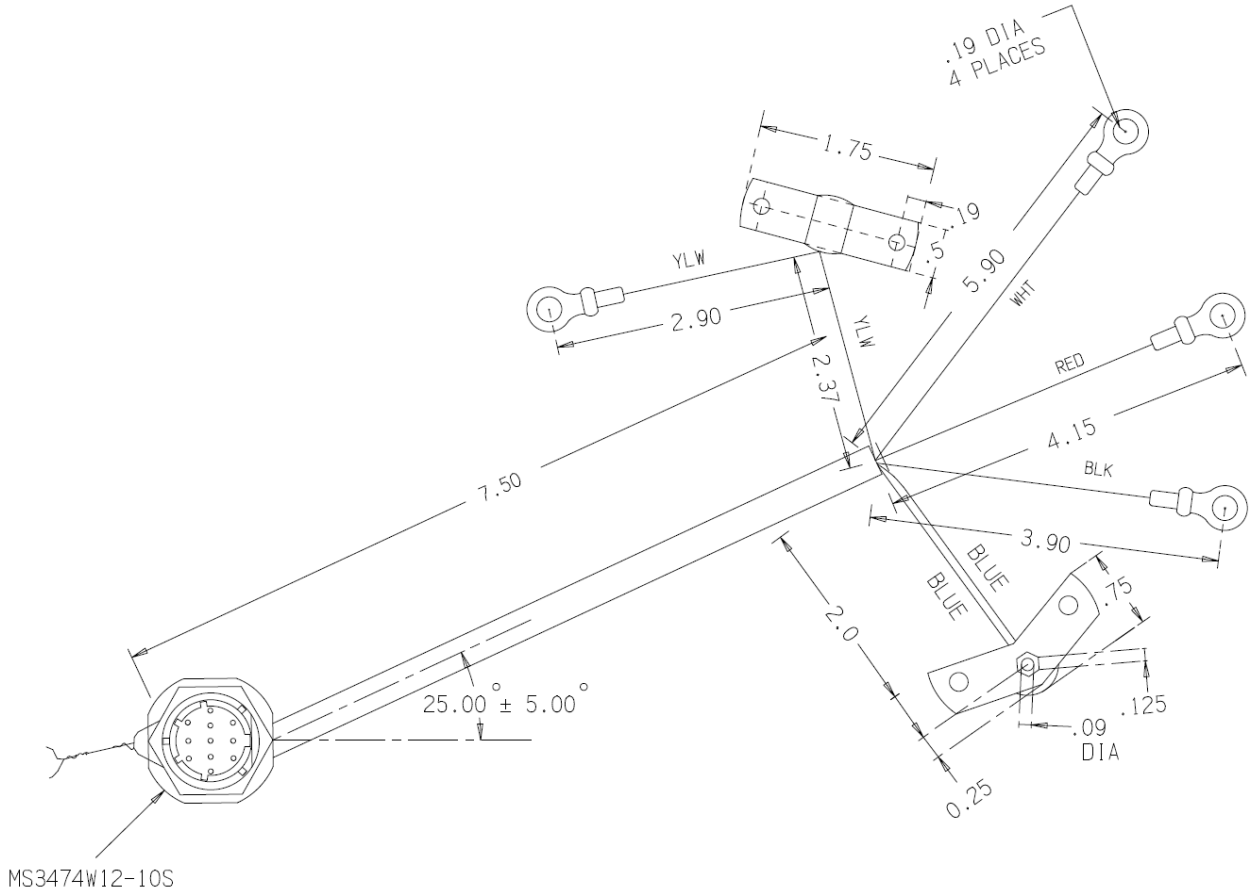
Table VIII. Group C inspection. Add the following:

Test No.	Examinations and tests	Requirement paragraph	Test paragraph
	Subgroup I		
I-10	Electrolyte resistance	3.9, 3.10.3	4.5.3
I-11	Temperature shock	3.9, 3.10.31	4.5.17
I-12	Vibration	3.9	4.5.16
	Subgroup II		
II-8	Wiring harness temperature test	3.10.31	4.5.33

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(side view)



(top view)

Figure 1: Wiring Harness, BB-432B/A, Side View and Top View

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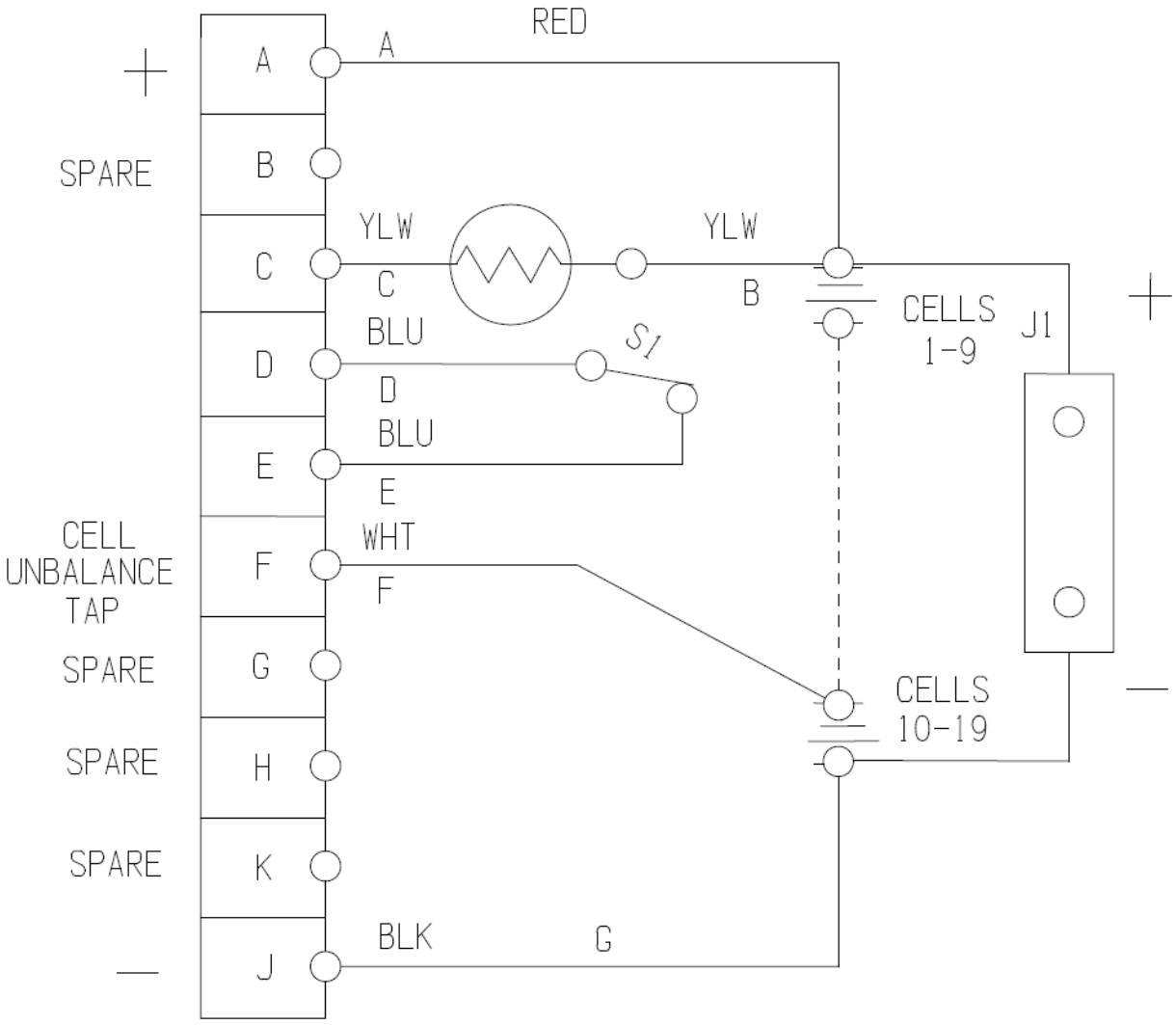


Figure 2: J2 connections

FIGURE NOTES:

Dimensions are in inches unless otherwise specified
All wire length tolerances ± .25 inches
Removable link tolerance: ± .015 inches

Custodians:
Army – CR

Preparing activity
Army – CR
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